

Topic model and topic codebook

```
tx <- list(
  seed = 2026L,
  min_docfreq = 5L, # DTM vocabulary floor
  run_searchK = FALSE, # TRUE for the long K-selection run
  K_grid = seq(5, 30, 3), # widen for a larger corpus
  K = 12L, # set from the searchK diagnostics
  spline_df = 5L,
  run_sensitivity = FALSE, # TRUE: refit STM at other parse floors
  sens_factors = c(0.5, 1.5), # floor multiples tested against the baseline
  facets_per_page = 9L, # topic trend panels per page
  workers = 16L,
  use_openrouter = TRUE, # FALSE to use the OpenAI-compatible endpoint
  # base_url_env and api_key_env are the NAMES of .Renviro variables, so the
  # small and large models can use different credentials and nothing
  # sensitive appears here. Ignored when use_openrouter = TRUE.
  steps = tibble::tribble(
    ~step, ~base_url_env, ~api_key_env, ~model,
    "label", "COMPASS_SMALL_URL", "COMPASS_SMALL_KEY", "MODEL-FOR-LABELING",
    "manager", "COMPASS_LARGE_URL", "COMPASS_LARGE_KEY", "MODEL-FOR-MANAGER")
  )

suppressPackageStartupMessages({
  library(here); library(stm); library(quanteda); library(tidytext)
  library(splines); library(furrr); library(viridis)
  library(tidyverse) # tidyverse LAST so its verbs win
})
# Used namespaces: clue (Hungarian matching), ellmer, glue.
theme_set(theme_minimal(base_size = 12))
options(knitr.table.max.print = Inf, knitr.kable.max_rows = Inf,
  max.print = 1e6)
set.seed(tx$seed)
`%||` <- function(x, y) if (is.null(x)) y else x
p_out <- function(f) here::here("outputs", f)
source(here::here("code", "llm_source.R"))
if (isTRUE(tx$use_openrouter)) {
  source(here::here("code", "llm_openrouter.R"))
  tx$steps$model <- open_router_models[tx$steps$step]
}
```

Input

The corpus is the paragraph units from 01_preprocess.qmd. Every cache below is fingerprinted against this file, so a changed parse cannot silently meet an old fit.

```

para_in <- p_out("paragraphs.csv")
sens_in <- p_out(c("paragraphs.csv", "blocks.csv", "linkage_plan.rds",
                  "block_embeddings.rds"))
invisible(fingerprint(para_in)) # stops by name if script 1 has not been rendered
paragraphs <- readr::read_csv(para_in, show_col_types = FALSE)
stopifnot("paragraphs.csv must carry the script 1 columns" =
          all(c("para_uid", "atom_id", "para_id", "Year", "n_tokens",
                "text") %in% names(paragraphs)))
writeLines(glue::glue("{nrow(paragraphs)} paragraph units from \\
                      {dplyr::n_distinct(paragraphs$atom_id)} documents."))

```

1481 paragraph units from 60 documents.

Tokenization

The DTM is built from cleaned tokens. Embeddings in script 1 used the raw text; this cleaning affects topics only.

```

# DTM cleaning (embeddings used the raw text; this affects topics only).
# Tokens must be purely alphabetic, which drops numbers, dates written in
# digits, and anything with special characters. Month and weekday names are
# stopped too: temporal words let topics encode Year directly, which would
# leak time into the prevalence model used to estimate trends over time.
temporal <- tibble(word = str_to_lower(c(month.name, month.abb,
    "monday", "tuesday", "wednesday", "thursday", "friday", "saturday",
    "sunday")))
tokens <- paragraphs |>
  dplyr::select(para_uid, text) |>
  tidytext::unnest_tokens(word, text) |>
  dplyr::mutate(word = str_remove_all(word, "'")) |>
  dplyr::filter(str_detect(word, "^[a-z]{2,}$")) |>
  dplyr::anti_join(tidytext::get_stopwords(), by = "word") |>
  dplyr::anti_join(temporal, by = "word")

```

Document-term matrix

Rare terms (under 5 paragraphs) are trimmed to stabilize estimation. meta is built from the surviving dfm rownames, so alignment with the model survives any dropped document. At work, check `nfeat(dfm_para)`: a vocabulary in the low thousands is comfortable at this scale, and a much larger one argues for a higher floor.

```

dfm_para <- tokens |>
  dplyr::count(para_uid, word) |>
  tidytext::cast_dfm(para_uid, word, n) |>
  quanteda::dfm_trim(min_docfreq = tx$min_docfreq)
stm_in <- quanteda::convert(dfm_para, to = "stm")
# as.data.frame, not a tibble: stm subsets metadata with base [ , ] semantics

```

```

# when building the prevalence model matrix, and a tibble returns a column
# rather than a vector there ("Error creating model matrix").
meta <- tibble(para_uid = rownames(dfm_para)) |>
  dplyr::left_join(dplyr::select(paragraphs, para_uid, atom_id, para_id, Year,
                                n_tokens, text), by = "para_uid") |>
  as.data.frame()
stopifnot("every dfm row must match a paragraph" = !anyNA(meta$Year),
  "meta must align with the stm documents" =
    nrow(meta) == length(stm_in$documents))

# Spline basis as ORDINARY COLUMNS, so no function call ever appears inside a
# formula. stm builds its prevalence model matrix by evaluating the formula
# against the metadata, and a call there (s(), ns(), or anything namespaced)
# is the documented cause of "Error creating model matrix". prev_form then
# reads ~ yr1 + yr2 + ..., which is also what estimateEffect uses below.
bs <- splines::ns(meta$Year, df = tx$spline_df)
colnames(bs) <- str_c("yr", seq_len(ncol(bs)))
meta <- cbind(meta, as.data.frame(bs))
prev_rhs <- str_c(colnames(bs), collapse = " + ") # "yr1 + yr2 + ..."
prev_form <- as.formula(str_c("~ ", prev_rhs))

```

Choosing K

stm's own parallelism forks, which is unavailable on Windows, so the held-out grid is parallelized explicitly with furrr. Read the panels together: take the smallest K where held-out likelihood has flattened and residual dispersion has stopped falling, then break ties on the coherence-exclusivity frontier. If the frontier is still improving at the top of the grid, widen tx\$K_grid rather than extrapolating. Semantic coherence is $\sum_{i < j} \log\{(D(v_i, v_j) + 1)/D(v_i)\}$ over each topic's top words, with the denominator the higher-ranked word.

```

if (isTRUE(tx$run_searchK)) {
  future::plan(future::multisession, workers = tx$workers)
  ho <- stm::make.heldout(stm_in$documents, stm_in$vocab, seed = tx$seed)
  metrics <- cache_guarded(p_out("searchK_metrics.csv"), para_in, {
    furrr::future_map(tx$K_grid, function(k) {
      fit <- stm::stm(ho$documents, ho$vocab, K = k, init.type = "Spectral",
        prevalence = prev_form, data = meta, verbose = FALSE)
      tibble(K = k,
        heldout = stm::eval.heldout(fit, ho$missing)$expected.heldout,
        dispersion = stm::checkResiduals(fit, ho$documents)$dispersion,
        coherence = mean(stm::semanticCoherence(fit, ho$documents)),
        exclusivity = mean(stm::exclusivity(fit)))
    }, .options = furrr::furrr_options(seed = tx$seed)) |> purrr::list_rbind()
  })
  future::plan(future::sequential)
  metrics |>
    tidyr::pivot_longer(-K) |>

```

```

ggplot(aes(K, value)) + geom_line(color = viridis(1, begin = 0.35)) +
  geom_point() + facet_wrap(~ name, scales = "free_y") + labs(y = NULL)
} else {
  if (is.null(tx$K))
    stop("tx$K is NULL and tx$run_searchK is FALSE. Re-run the config chunk, ",
         "or set tx$K before continuing.")
  glue::glue("searchK skipped (tx$run_searchK = FALSE); K fixed at {tx$K}")
}

```

```
searchK skipped (tx$run_searchK = FALSE); K fixed at 12
```

Structural topic model

Year enters the prevalence model as a spline (df 5), so topic attention is allowed to drift over the period instead of being averaged away; spectral initialization is deterministic given the data, removing initialization luck from a fit everything downstream depends on.

```

stm_fit <- cache_guarded(p_out("stm_fit.rds"), para_in,
  stm::stm(stm_in$documents, stm_in$vocab, K = tx$K, init.type = "Spectral",
    prevalence = prev_form, data = meta, verbose = FALSE))

# K comes from the FITTED model, not the config: a cached fit can predate a
# change to tx$K, and every downstream object is sized by the fit.
K_fit <- ncol(stm_fit$theta)
if (K_fit != tx$K)
  stop("Cached stm_fit.rds has K = ", K_fit, " but tx$K = ", tx$K,
       ". Delete outputs/stm_fit.rds (and stm_year_effects.rds, ",
       "topic_trends.csv) once, then re-render.")

theta <- stm_fit$theta
colnames(theta) <- str_c("topic_", seq_len(K_fit))
paragraph_theta <- meta |> dplyr::bind_cols(tibble::as_tibble(theta))

```

Modal topic assignment and its diagnostics

Each paragraph is assigned its **modal** topic, so topics partition the corpus (mutually exclusive, collectively exhaustive) and each paragraph belongs to exactly one topic's frame in 03_stance.qmd. This is the same move as modal class assignment in latent class analysis, and it inherits the same caveat: a modal assignment is an estimate, not an observation, and treating it as observed pushes classification error downstream. Two LCA diagnostics report how much error to expect. **Entropy** R^2 summarizes overall assignment sharpness on a 0-1 scale; **AvePP**, the average posterior probability among a topic's own assigned paragraphs, does it per topic, with about 0.7 the conventional floor. Unlike LCA, STM supplies no classification error matrix over modal assignment, so no BCH-style correction is available here; the honest substitute is the min_theta scope floor in 03_stance.qmd, which drops paragraphs that only weakly belong to their modal topic.

```

paragraph_theta <- paragraph_theta |>
  dplyr::mutate(topic = max.col(theta, ties.method = "first"),
               theta_max = theta[cbind(dplyr::row_number(), topic)])

ent <- -sum(theta * log(theta + 1e-12))
entropy_r2 <- 1 - ent / (nrow(theta) * log(K_fit))

topic_summary <- tibble(
  topic = seq_len(K_fit),
  prevalence = colMeans(theta),
  frex = apply(stm::labelTopics(stm_fit, n = 8)$frex, 1, str_c,
               collapse = ", "),
  prob = apply(stm::labelTopics(stm_fit, n = 8)$prob, 1, str_c,
               collapse = ", ")) |>
  dplyr::left_join(
    paragraph_theta |>
      dplyr::group_by(topic) |>
      dplyr::summarise(n_paras = dplyr::n(), avepp = mean(theta_max)),
    by = "topic")

readr::write_csv(topic_summary, p_out("topic_summary.csv"))
readr::write_csv(paragraph_theta, p_out("paragraph_theta.csv"))
theta_in <- p_out("paragraph_theta.csv")

glue::glue("Entropy R2 = {round(entropy_r2, 3)}; topics with AvePP < 0.7: ",
           "{sum(topic_summary$avepp < 0.7)} of {K_fit}")

```

Entropy R2 = 0.477; topics with AvePP < 0.7: 12 of 12

```

topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::select(topic, prevalence, n_paras, avepp) |>
  tbl_paged(caption = "Topics by prevalence, with assigned paragraph counts and
AvePP (below 0.7 means fuzzy modal assignment).")

```

Topics by prevalence, with assigned paragraph counts and AvePP (below 0.7 means fuzzy modal assignment).

topic	prevalence	n_paras	avepp
6	0.166	301	0.553
8	0.108	151	0.550
3	0.101	126	0.510
2	0.099	136	0.562
5	0.091	126	0.561

topic	prevalence	n_paras	avepp
1	0.081	103	0.517
4	0.068	114	0.574
7	0.068	106	0.572
9	0.060	88	0.583
11	0.059	82	0.546
10	0.059	93	0.621
12	0.043	55	0.517

```
topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::mutate(ttl = str_c("Topic ", topic)) |>
  tbl_records("ttl", c("frex", "prob"),
    caption = "Topic terms: the evidence for checking each label.")
```

Topic terms: the evidence for checking each label.

Topic 6 - frex: americans, americas, together, story, let, journey, heart, america - prob: us, america, let, can, must, americans, world, together

Topic 8 - frex: health, back, across, vice, voices, mr, century, moment - prob: new, us, years, now, people, today, time, government

Topic 3 - frex: problems, economic, opportunity, ideals, concern, self, civilization, education - prob: people, must, american, government, free, freedom, world, liberty

Topic 2 - frex: god, oath, thank, day, bless, president, defend, presence - prob: god, day, people, president, united, citizens, oath, nation

Topic 5 - frex: peace, international, nations, arms, practice, trade, settlement, effort - prob: peace, nations, world, can, shall, must, us, war

Topic 1 - frex: little, pride, women, men, matters, helped, enterprise, face - prob: men, national, can, good, great, life, upon, nation

Topic 4 - frex: laws, amendment, law, party, enforce, department, negro, suffrage - prob: laws, law, government, public, constitution, party, shall, people

Topic 7 - frex: revenue, expenditures, business, currency, taxation, debt, money, credit - prob: government, public, people, revenue, business, upon, can, power

Topic 9 - frex: enforcement, discharge, duties, station, providence, gratitude, enter, press - prob: shall, duties, fellow, public, citizens, country, upon, confidence

Topic 11 - frex: meaning, sentiment, relations, cultivate, british, happy, friendly, peculiar - prob: government, states, people, great, war, united, every, rights

Topic 10 - frex: territory, confederacy, extended, union, intercourse, texas, twenty, population - prob: states, union, government, constitution, people, upon, powers, one

Topic 12 - frex: congress, navy, naval, pursued, army, session, improvement, coast - prob: congress, states, great, executive, war, force, power, upon

Does the parse drive the topics?

The floor is calibrated to the corpus, but the floor is still a decision about the document unit, and the topic model is sensitive to document length. This section tests whether that choice matters. The unparagraphed documents are relinked at 0.5 and 1.5 times the floor calibrated in script 1, using the vectors cached there, so no API calls are made, the STM is refit at the same K on each, and every baseline topic is matched one-to-one to its best counterpart in each alternative fit by the cosine of their topic-word distributions over the shared vocabulary (Hungarian assignment via `clue::solve_LSAP`).

The result is reported as similarities, not as a pass or fail, since any threshold would be invented. Read it this way: if most matched cosines are high at both alternative floors, topics are properties of the corpus and not of the parse; if a few topics match poorly, those are the ones whose existence depends on how text was grouped, and they deserve caution before being carried into stance measurement.

```
if (isTRUE(tx$run_sensitivity)) {
  blocks <- readr::read_csv(p_out("blocks.csv"), show_col_types = FALSE)
  plan <- readRDS(p_out("linkage_plan.rds"))
  emb <- readRDS(p_out("block_embeddings.rds"))
  if (!any(!plan$doc$paragraphed))
    writeLines("No documents were relinked in script 1; the test has nothing to vary.")
  refit <- function(f) {
    p <- link_corpus(blocks, emb$vec, plan$floor * f, plan$doc)
    d <- p |>
      dplyr::select(para_uid, text) |>
      tidytext::unnest_tokens(word, text) |>
      dplyr::mutate(word = str_remove_all(word, "'")) |>
      dplyr::filter(str_detect(word, "^[a-z]{2,}$")) |>
      dplyr::anti_join(tidytext::get_stopwords(), by = "word") |>
      dplyr::anti_join(temporal, by = "word") |>
      dplyr::count(para_uid, word) |>
      tidytext::cast_dfm(para_uid, word, n) |>
      quantda::dfm_trim(min_docfreq = tx$min_docfreq)
    s_in <- quantda::convert(d, to = "stm")
    list(n_docs = length(s_in$documents),
         fit = stm::stm(s_in$documents, s_in$vocab, K = K_fit,
                        init.type = "Spectral", verbose = FALSE))
  }
  sens <- cache_guarded(p_out("sensitivity_fits.rds"), sens_in,
```

```

        purrr::map(tx$sens_factors, refit) |>
        purrr::set_names(tx$sens_factors))
b_ref <- beta_matrix(stm_fit)
sens_tbl <- purrr::imap(sens, function(x, f)
  match_topics(b_ref, beta_matrix(x$fit)) |>
  dplyr::mutate(floor_factor = as.numeric(f), n_docs = x$n_docs)) |>
  purrr::list_rbind()
readr::write_csv(sens_tbl, p_out("parse_sensitivity.csv"))

sens_tbl |>
  dplyr::group_by(floor_factor) |>
  dplyr::summarise(documents = dplyr::first(n_docs),
    median_cosine = stats::median(cosine),
    lowest_cosine = min(cosine),
    shared_vocab = dplyr::first(shared_vocab),
    .groups = "drop") |>
  tbl_paged(caption = "Topic stability across parse floors: matched topic-word
cosine against the baseline fit.")
sens_tbl |>
  tidyr::pivot_wider(id_cols = topic, names_from = floor_factor,
    values_from = cosine, names_prefix = "floor_x") |>
  dplyr::left_join(dplyr::select(topic_summary, topic, prevalence),
    by = "topic") |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  tbl_paged(caption = "Per-topic matched cosine by floor multiple. Low values
mark topics whose existence depends on the parse.")
} else {
  writeLines("Sensitivity test skipped (tx$run_sensitivity = FALSE).")
}

```

Sensitivity test skipped (tx\$run_sensitivity = FALSE).

Topic codebook

The codebook is written **once**, after the final STM, and is the naming record from here on. If `topic_codebook.csv` already exists on disk, this section reads it and makes no API calls: the file is assumed to carry your edits, and re-drafting would overwrite them. Delete the file to redraft from scratch.

Two passes, deliberately shaped:

Pass 1, one call per topic with the smaller model (label step). Each call sees only that topic's own evidence: its FREX terms, its highest-probability terms, and its three highest-theta paragraphs. It is blind to every other topic, to Year, and to stance. That blindness is the point: a model shown all topics at once starts fitting a narrative across them, and labels stop being traceable to the evidence a human can check them against.

Pass 2, one manager call with the larger model (manager step). Blind drafting has one predictable failure: two topics can come back with near-duplicate labels even though their term distributions differ. The manager sees all K labels with their FREX terms and resolves collisions. Its mandate is deliberately narrow, stated in the prompt and enforced by the schema: it may **rename only**. It cannot merge topics, split them, change a description, or reorder anything, because the topic structure is the statistical model's output and an LLM should not be permitted to override it.

The proposition column is written empty here. 03_stance.qmd fills it for the one topic it processes, since propositions are stance targets and you only need them for the topics you actually analyze.

```
cb_path <- p_out("topic_codebook.csv")

# A refit can renumber topics, and a codebook written for the old fit would
# then label the wrong topics silently. The codebook's fingerprint is taken on
# the theta it was drafted from; your edits to the CSV itself never trip it.
check_fingerprint(cb_path, theta_in, strict = FALSE)
if (file.exists(cb_path)) {
  codebook <- readr::read_csv(cb_path, show_col_types = FALSE)
  writeLines(glue::glue("Existing codebook found ({nrow(codebook)} topics); \\
                        no LLM calls made. Delete {basename(cb_path)} to \\
                        redraft."))
} else {
  ev <- topic_summary |>
    dplyr::left_join(
      paragraph_theta |>
        dplyr::group_by(topic) |>
        dplyr::slice_max(theta_max, n = 3) |>
        dplyr::summarise(ex = str_c(str_trunc(text, 300), collapse = "\n- "),
                        .groups = "drop"),
      by = "topic")

  # ---- Pass 1: one blind call per topic -----
  label_sys <- build_prompt(
    role = str_c("You are a survey methodologist who names statistical topics",
      ",
      "so colleagues can use them."),
    task = str_c("Read this topic's distinctive terms and example paragraphs.",
      ",
      "Write a short label and one factual sentence describing what",
      ",
      "these paragraphs are about. Before answering, check that ",
      "every word of your description is supported by the evidence ",
      "shown."),
    rules = c("Use only the terms and paragraphs shown.",
      "Do not infer stance, sentiment, intent, or who is speaking.",
      "Do not invent statistics, dates, places, or named entities.",
      "If the evidence is mixed, describe the mixture plainly."),
```

```

output = "A label of at most 4 words and a one-sentence description.")
label_schema <- ellmer::type_object(
  label = ellmer::type_string("Topic name, at most 4 words."),
  description = ellmer::type_string("One factual sentence.")
)
label_chat <- make_chat("label", label_sys, tx$steps, tx$use_openrouter)

drafts <- purrr::map(seq_len(nrow(ev)), function(i) {
  r <- call_one(label_chat,
    str_c("Distinctive (FREX) terms: ", ev$frex[i],
      "\nHighest-probability terms: ", ev$prob[i],
      "\nExample paragraphs:\n- ", ev$ex[i]), label_schema)
  tibble(topic = ev$topic[i],
    label = r$label %||% NA_character_,
    description = r$description %||% NA_character_)
}) |> purrr::list_rbind()
if (anyNA(drafts$label))
  warning(sum(is.na(drafts$label)), " topic label calls failed; those rows ",
    "are blank in the codebook and need a name by hand.")

# ---- Pass 2: manager resolves duplicate labels, renaming ONLY -----
mgr_sys <- build_prompt(
  role = str_c("You are a senior methodologist reviewing a topic codebook ",
    "for clarity before it goes to coders."),
  task = str_c("You are shown every topic's draft label with its ",
    "distinctive terms. Some labels are near-duplicates of each ",
    "other even though the topics differ. Return a label for ",
    "EVERY topic: unchanged where it is already distinct, and ",
    "revised only where two or more labels would be confused. ",
    "Before answering, check that no two returned labels are ",
    "the same and that every topic number appears exactly once."),
  rules = c("Rename only. Never merge, split, drop, or reorder topics.",
    "Return the same number of topics you were given.",
    "Base any rename on the distinctive terms shown, nothing else.",
    "Keep labels to at most 4 words.",
    "Do not write descriptions; they are not yours to change."),
  output = "One label per topic number.")
mgr_schema <- ellmer::type_object(
  topics = ellmer::type_array(
    items = ellmer::type_object(
      topic = ellmer::type_integer("The topic number, unchanged."),
      label = ellmer::type_string("Final label, at most 4 words.)))
mgr_prompt <- str_c(
  "Draft labels with their distinctive terms:\n",
  str_c("- topic ", drafts$topic, ": \"", drafts$label, "\" | terms: ",
    ev$frex[match(drafts$topic, ev$topic)], collapse = "\n"))
mgr <- call_one(make_chat("manager", mgr_sys, tx$steps, tx$use_openrouter),
  mgr_prompt, mgr_schema)

```

```

# The manager is advisory: its output is applied ONLY if it returns exactly
# one label per topic with no topics invented or lost. Any deviation and the
# blind drafts stand, because a malformed rename must never reshape the
# codebook silently. ellmer converts an array of objects into a data frame,
# so both shapes are accepted; checking length() of a data frame counts its
# columns, which is how a valid response can be rejected without anyone
# noticing. The status line below says which path was taken.
mgr_tbl <- if (is.null(mgr)) NULL else if (is.data.frame(mgr$topics))
  tibble::as_tibble(mgr$topics) else
  purrr::map(mgr$topics, ~ tibble(topic = .x$topic, label = .x$label)) |>
  purrr::list_rbind()
mgr_ok <- !is.null(mgr_tbl) && nrow(mgr_tbl) == nrow(drafts) &&
  setequal(as.integer(mgr_tbl$topic), drafts$topic)
mgr_status <- if (is.null(mgr_tbl)) "no response; blind drafts kept" else
  if (!mgr_ok) glue::glue("rejected: returned {nrow(mgr_tbl)} labels for \\
    {nrow(drafts)} topics, or different topic numbers; blind drafts kept")
else
  "applied"
final <- if (mgr_ok) {
  drafts |>
    dplyr::left_join(dplyr::transmute(mgr_tbl, topic = as.integer(topic),
                                      label_new = label), by = "topic") |>
    dplyr::mutate(label = dplyr::coalesce(label_new, label)) |>
    dplyr::select(-label_new)
} else drafts
n_renamed <- sum(final$label != drafts$label, na.rm = TRUE)

codebook <- final |>
  dplyr::mutate(proposition = NA_character_) |> # 03_stance.qmd fills this
  dplyr::left_join(topic_summary |> dplyr::select(topic, prevalence, avepp),
    by = "topic") |>
  dplyr::arrange(topic)
readr::write_csv(codebook, cb_path)
write_fingerprint(cb_path, theta_in)
writeLines(glue::glue("Codebook drafted: {nrow(codebook)} topics. Manager \\
  pass: {mgr_status}; {n_renamed} label(s) renamed. EDIT \\
  {basename(cb_path)} before running 03_stance.qmd; it will not be \\
  redrafted while the file exists."))
}

```

Codebook drafted: 12 topics. Manager pass: applied; 5 label(s) renamed. EDIT topic_codebook.csv before running 03_stance.qmd; it will not be redrafted while the file exists.

```

# Duplicate labels are checked on every render, drafted or hand-edited: two
# topics sharing a name are indistinguishable in every table, plot, and in the
# dashboard, and a coder cannot tell which one a paragraph belongs to.
dup <- codebook |>

```

```

dplyr::mutate(key = str_to_lower(str_squish(label))) |>
dplyr::group_by(key) |>
dplyr::filter(dplyr::n() > 1) |>
dplyr::ungroup()
if (nrow(dup)) {
  warning("Duplicate topic labels: ", str_c(unique(dup$label), collapse = "; "),
    ". Rename them in ", basename(cb_path), ".")
  writelines(glue::glue("**Duplicate labels to fix:** \\
    {str_c('topic ', dup$topic, ' (', dup$label, ')', collapse = ', ')} \\
    Rename them in {basename(cb_path)} using each topic's terms above.\\
  "))
}

codebook |>
dplyr::mutate(ttl = str_c("Topic ", topic, ": ", label)) |>
tbl_records("ttl", "description",
  caption = "Topic codebook. Labels drafted blind per topic, then
deduplicated by a rename-only manager pass. Edit the CSV; nothing downstream
redrafts it.")

```

Topic codebook. Labels drafted blind per topic, then deduplicated by a rename-only manager pass. Edit the CSV; nothing downstream redrafts it.

Topic 1: National Enterprise - description: The paragraphs discuss the need to conserve national resources and restore the balance of population in industrial centers to support the nation's enterprise and life.

Topic 2: Oath of Office - description: These paragraphs are about a president taking the oath of office and expressing gratitude and commitment to leading the nation.

Topic 3: American Democracy - description: These paragraphs discuss the importance of education, self-government, and individual freedom in the context of American democracy and its ideals.

Topic 4: Government Laws - description: These paragraphs are about the administration and enforcement of laws by the government.

Topic 5: International Peace - description: The paragraphs discuss efforts to restore and maintain peace and stability in the world through international relations and diplomatic consultations.

Topic 6: American Society - description: These paragraphs are about building an American opportunity society where all Americans can go forward together.

Topic 7: Government Finance - description: These paragraphs are about government revenue, expenditures, taxation, and debt.

Topic 8: Presidential Speech - description: These paragraphs are about government and the state of the people across the country today.

Topic 9: Presidential Duties - description: These paragraphs are about a person assuming a high public office and expressing gratitude for the confidence shown by their fellow-citizens.

Topic 10: US Territory - description: The paragraphs discuss the structure and powers of the US government, including its departments, authorities, and the relationship between states and the union.

Topic 11: US Foreign Relations - description: The paragraphs discuss the United States Government and its relation to the nation and its people.

Topic 12: US Military Power - description: These paragraphs are about the power of the Executive and the role of the navy and other military forces in the United States.

Representative paragraphs

```
top4 <- topic_summary |> dplyr::slice_max(prevalence, n = 4) |>
  dplyr::pull(topic)
th <- stm::findThoughts(stm_fit, texts = meta$para_uid, n = 3, topics = top4)
tibble(topic = rep(top4, each = 3), para_uid = unlist(th$docs)) |>
  dplyr::left_join(paragraphs |> dplyr::select(para_uid, text),
    by = "para_uid") |>
  dplyr::mutate(ttl = str_c("Topic ", topic, " | ", para_uid)) |>
  tbl_records("ttl", "text",
    caption = "Three highest-theta paragraphs for the four most
prevalent topics.")
```

Three highest-theta paragraphs for the four most prevalent topics.

Topic 6 | 1985-Reagan-28 - text: Let us resolve that we the people will build an American opportunity society in which all of us - white and black, rich and poor, young and old - will go forward together arm in arm. Again, let us remember that though our heritage is one of blood lines from every corner of the Earth, we are all Americans pledged to carry on this last, best hope of man on Earth.

Topic 6 | 1997-Clinton-29 - text: America demands and deserves big things from us - and nothing big ever came from being small. Let us remember the timeless wisdom of Cardinal Bernardin, when facing the end of his own life. He said: "It is wrong to waste the precious gift of time, on acrimony and division."

Topic 6 | 1965-Johnson-10 - text: In a land of great wealth, families must not live in hopeless poverty. In a land rich in harvest, children just must not go hungry. In a land of healing miracles, neighbors must not suffer and die unattended. In a great land of learning and scholars, young people must be taught to read and write.

Topic 8 | 1989-Bush-20 - text: To my friends - and yes, I do mean friends - in the loyal opposition - and yes, I mean loyal: I put out my hand. I am putting out my hand to you, Mr. Speaker. I am putting out my hand to you Mr. Majority Leader. For this is the thing: This is the age of the offered hand. We can't turn back clocks, and I don't want to. But when our fathers were young, Mr. Speaker, our differences ended at the water's edge. And we don't wish to turn back time, but when our mothers were young, Mr. Majority Leader, the Congress and the Executive were capable of working together to produce a budget on which this nation could live. Let us negotiate soon and

hard. But in the end, let us produce. The American people await action. They didn't send us here to bicker. They ask us to rise above the merely partisan. "In crucial things, unity" - and this, my friends, is crucial.

Topic 8 | 1969-Nixon-10 - text: Our crisis today is the reverse. We have found ourselves rich in goods, but ragged in spirit; reaching with magnificent precision for the moon, but falling into raucous discord on earth. We are caught in war, wanting peace. We are torn by division, wanting unity. We see around us empty lives, wanting fulfillment. We see tasks that need doing, waiting for hands to do them. To a crisis of the spirit, we need an answer of the spirit. To find that answer, we need only look within ourselves. When we listen to "the better angels of our nature," we find that they celebrate the simple things, the basic things - such as goodness, decency, love, kindness. Greatness comes in simple trappings. The simple things are the ones most needed today if we are to surmount what divides us, and cement what unites us. To lower our voices would be a simple thing.

Topic 8 | 2009-Obama-19 - text: What the cynics fail to understand is that the ground has shifted beneath them - that the stale political arguments that have consumed us for so long no longer apply. The question we ask today is not whether our government is too big or too small, but whether it works - whether it helps families find jobs at a decent wage, care they can afford, a retirement that is dignified. Where the answer is yes, we intend to move forward. Where the answer is no, programs will end. And those of us who manage the public's dollars will be held to account - to spend wisely, reform bad habits, and do our business in the light of day - because only then can we restore the vital trust between a people and their government.

Topic 3 | 1929-Hoover-24 - text: Although education is primarily a responsibility of the States and local communities, and rightly so, yet the Nation as a whole is vitally concerned in its development everywhere to the highest standards and to complete universality. Self-government can succeed only through an instructed electorate. Our objective is not simply to overcome illiteracy. The Nation has marched far beyond that. The more complex the problems of the Nation become, the greater is the need for more and more advanced instruction. Moreover, as our numbers increase and as our life expands with science and invention, we must discover more and more leaders for every walk of life. We can not hope to succeed in directing this increasingly complex civilization unless we can draw all the talent of leadership from the whole people. One civilization after another has been wrecked upon the attempt to secure sufficient leadership from a single group or class. If we would prevent the growth of class distinctions and would constantly refresh our leadership with the ideals of our people, we must draw constantly from the general mass. The full opportunity for every boy and girl to rise through the selective processes of education can alone secure to us this leadership.

Topic 3 | 1921-Harding-02 - text: When one surveys the world about him after the great storm, noting the marks of destruction and yet rejoicing in the ruggedness of the things which withstood it, if he is an American he breathes the clarified atmosphere with a strange mingling of regret and new hope. We have seen a world passion spend its fury, but we contemplate our Republic unshaken, and hold our civilization secure. Liberty - liberty within the law - and civilization are inseparable, and though both were threatened we find them now secure; and there comes to

Americans the profound assurance that our representative government is the highest expression and surest guaranty of both.

Topic 3 | 1949-Truman-09 - text: Communism is based on the belief that man is so weak and inadequate that he is unable to govern himself, and therefore requires the rule of strong masters. Democracy is based on the conviction that man has the moral and intellectual capacity, as well as the inalienable right, to govern himself with reason and justice.

Topic 2 | 2009-Obama-02 - text: I stand here today humbled by the task before us, grateful for the trust you have bestowed, mindful of the sacrifices borne by our ancestors. I thank President Bush for his service to our nation, as well as the generosity and cooperation he has shown throughout this transition.

Topic 2 | 1937-Roosevelt-36 - text: In taking again the oath of office as President of the United States, I assume the solemn obligation of leading the American people forward along the road over which they have chosen to advance.

Topic 2 | 1921-Harding-34 - text: I have taken the solemn oath of office on that passage of Holy Writ wherein it is asked: "What doth the Lord require of thee but to do justly, and to love mercy, and to walk humbly with thy God?" This I plight to God and country.

Prevalence over time

```
stm_year_effects <- cache_guarded(p_out("stm_year_effects.rds"), para_in,
  stm::estimateEffect(
    as.formula(str_c("1:", K_fit, " ~ ", prev_rhs)),
    stm_fit, metadata = meta, uncertainty = "Global"))
```

Expected topic prevalence is predicted across the observed year range from the fitted effect object, with 95 percent intervals. Two details make the prediction correct rather than merely plausible. The spline basis is rebuilt at the prediction grid with `predict(bs, ...)`, which reuses the **original knots**; calling `ns()` afresh on the grid would place knots differently and mis-predict silently. Curves can dip slightly below zero where a topic is nearly absent: `estimateEffect` fits a linear model to topic proportions, which does not respect the 0 bound, so read values below zero as zero rather than as an error. And the intervals come from the simulation draws `estimateEffect` already stores under `uncertainty = "Global"`, so they carry the topic-model uncertainty rather than regression standard errors alone.

```
# Prediction grid: every observed year, with the basis evaluated at the
# ORIGINAL knots so the fitted coefficients apply.
grid <- tibble(Year = sort(unique(meta$Year)))
Xg    <- cbind(1, predict(bs, newx = grid$Year))

# estimateEffect stores per-topic simulation draws of the coefficient vector
# in $parameters[[k]][[j]]$est. Predicting each draw and taking quantiles
# propagates both the regression and the topic-model uncertainty.
```

```

draws_for_topic <- function(k) {
  purrr::map(stm_year_effects$parameters[[k]], function(sim)
    as.numeric(Xg %*% sim$est)) |>
  purrr::reduce(rbind) |> matrix(ncol = nrow(grid))
}
topic_trends <- purrr::map(seq_along(stm_year_effects$topics), function(i) {
  d <- draws_for_topic(i)
  grid |> dplyr::mutate(
    topic = stm_year_effects$topics[i],
    fit   = apply(d, 2, mean),
    lo    = apply(d, 2, quantile, 0.025),
    hi    = apply(d, 2, quantile, 0.975))
}) |> purrr::list_rbind()
readr::write_csv(topic_trends, p_out("topic_trends.csv"))
glue::glue("Predicted prevalence for {dplyr::n_distinct(topic_trends$topic)} \
\
      topics across {nrow(grid)} years written to topic_trends.csv")

```

Predicted prevalence for 12 topics across 60 years written to topic_trends.csv

```

# All topics, paged so panels stay legible: 25 facets on one page renders each
# about 1.5 inches wide and the ribbons become unreadable.
trend_panel <- function(d, page, n_pages) {
  ggplot(d, aes(Year, fit)) +
    geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2,
              fill = viridis(1, begin = 0.35)) +
    geom_line(linewidth = 0.7, color = viridis(1, begin = 0.35)) +
    facet_wrap(~ topic_label, scales = "free_y", ncol = 3) +
    labs(x = NULL, y = "expected topic proportion",
         title = glue::glue("Topic prevalence over time ({page} of {n_pages})"),
         subtitle = "Model-based estimates with 95 percent intervals")
}
trend_pages <- topic_trends |>
  dplyr::left_join(topic_summary |> dplyr::select(topic, prevalence),
                  by = "topic") |>
  dplyr::left_join(codebook |> dplyr::select(topic, label), by = "topic") |>
  dplyr::mutate(
    topic_label = str_c("T", topic, ":", str_trunc(label, 22)),
    page = (dplyr::dense_rank(dplyr::desc(prevalence)) - 1L) %/%
            tx$facets_per_page + 1L) |>
  dplyr::group_split(page)
purrr::iwalk(trend_pages, function(d, i)
  print(trend_panel(d, i, length(trend_pages))))

```

Topic prevalence over time (1 of 2)
Model-based estimates with 95 percent intervals

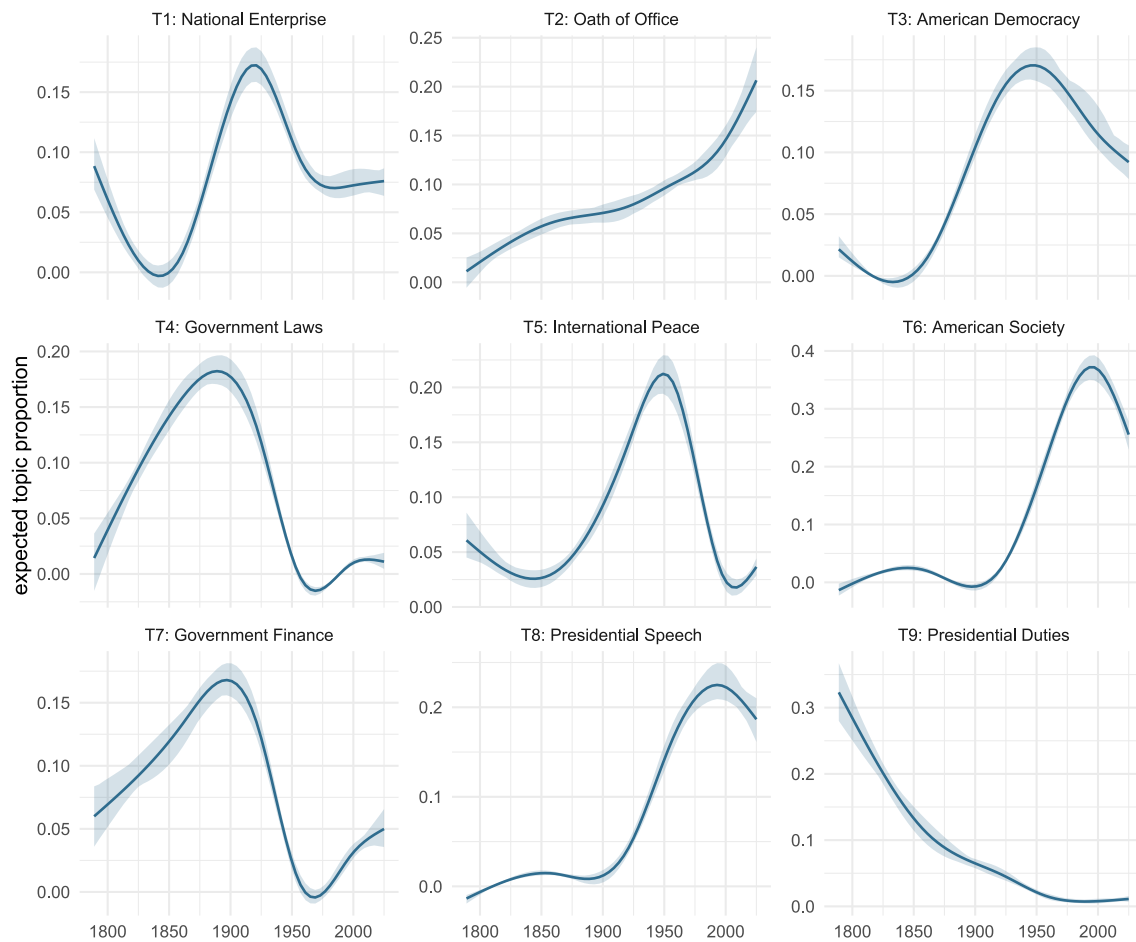


Figure 1: Expected topic proportion by year with 95 percent intervals, from estimateEffect. The model is linear in proportions, so stretches where a topic is nearly absent can dip below zero; read those as zero.

Topic prevalence over time (2 of 2)
Model-based estimates with 95 percent intervals

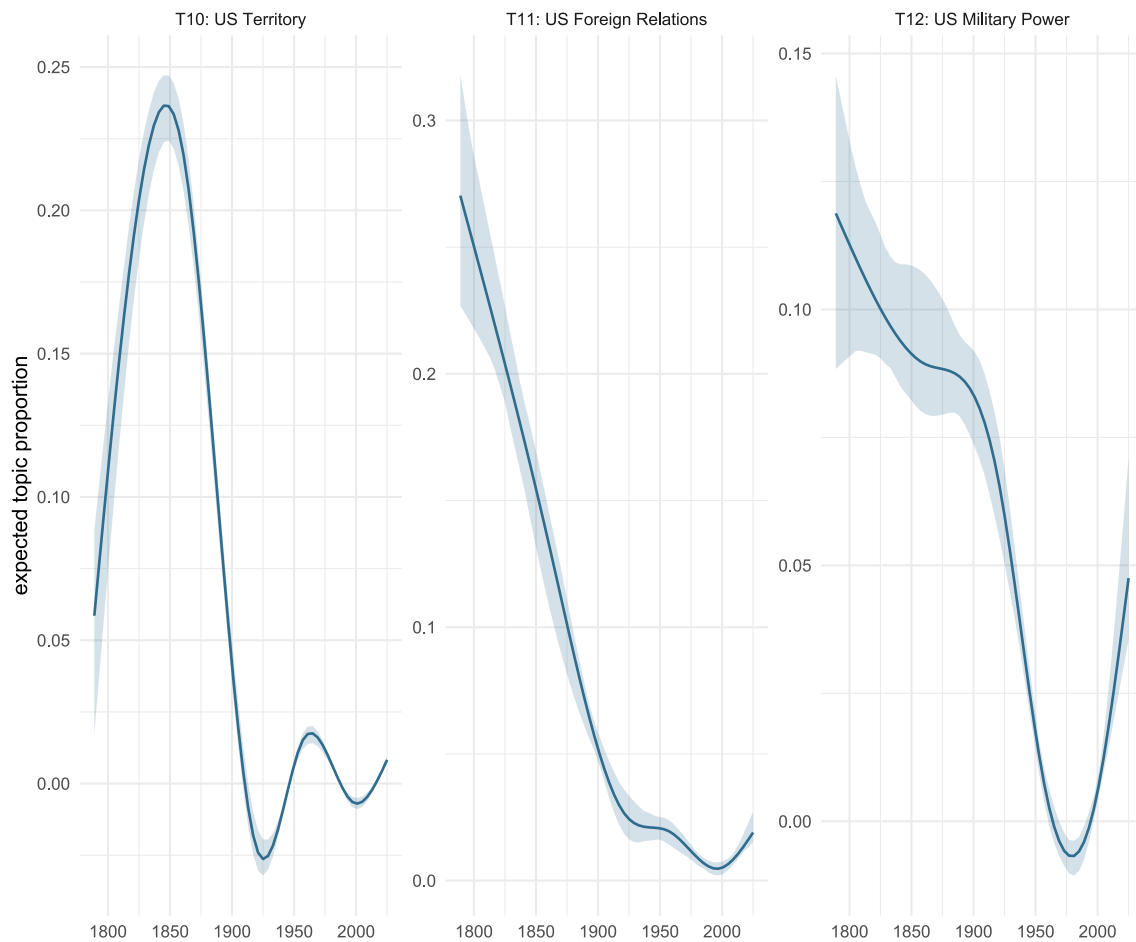


Figure 2: Expected topic proportion by year with 95 percent intervals, from estimateEffect. The model is linear in proportions, so stretches where a topic is nearly absent can dip below zero; read those as zero.

```
glue::glue("outputs/ written: paragraphs.csv, paragraph_theta.csv (ids, Year, \
\
n_tokens, text, topic_1..K, topic, theta_max), topic_summary.csv, \
topic_codebook.csv (labels; 03_stance.qmd adds propositions), topic_trends.csv, \
stm_fit.rds, stm_year_effects.rds. 03_stance.qmd reads these.")
```

```
outputs/ written: paragraphs.csv, paragraph_theta.csv (ids, Year, n_tokens,
text, topic_1..K, topic, theta_max), topic_summary.csv, topic_codebook.csv
(labels; 03_stance.qmd adds propositions), topic_trends.csv,
stm_fit.rds, stm_year_effects.rds. 03_stance.qmd reads these.
```